
ECE510 Post-silicon Validation

DFX

Outline

- Type of tests
- DFX overview
 - What
 - Coverage
 - Fault model
- DFX types
 - ATPG, ALPG, BIST
 - Scanout, Scan
 - Functional
 - Redundancy
 - LYA
 - Others

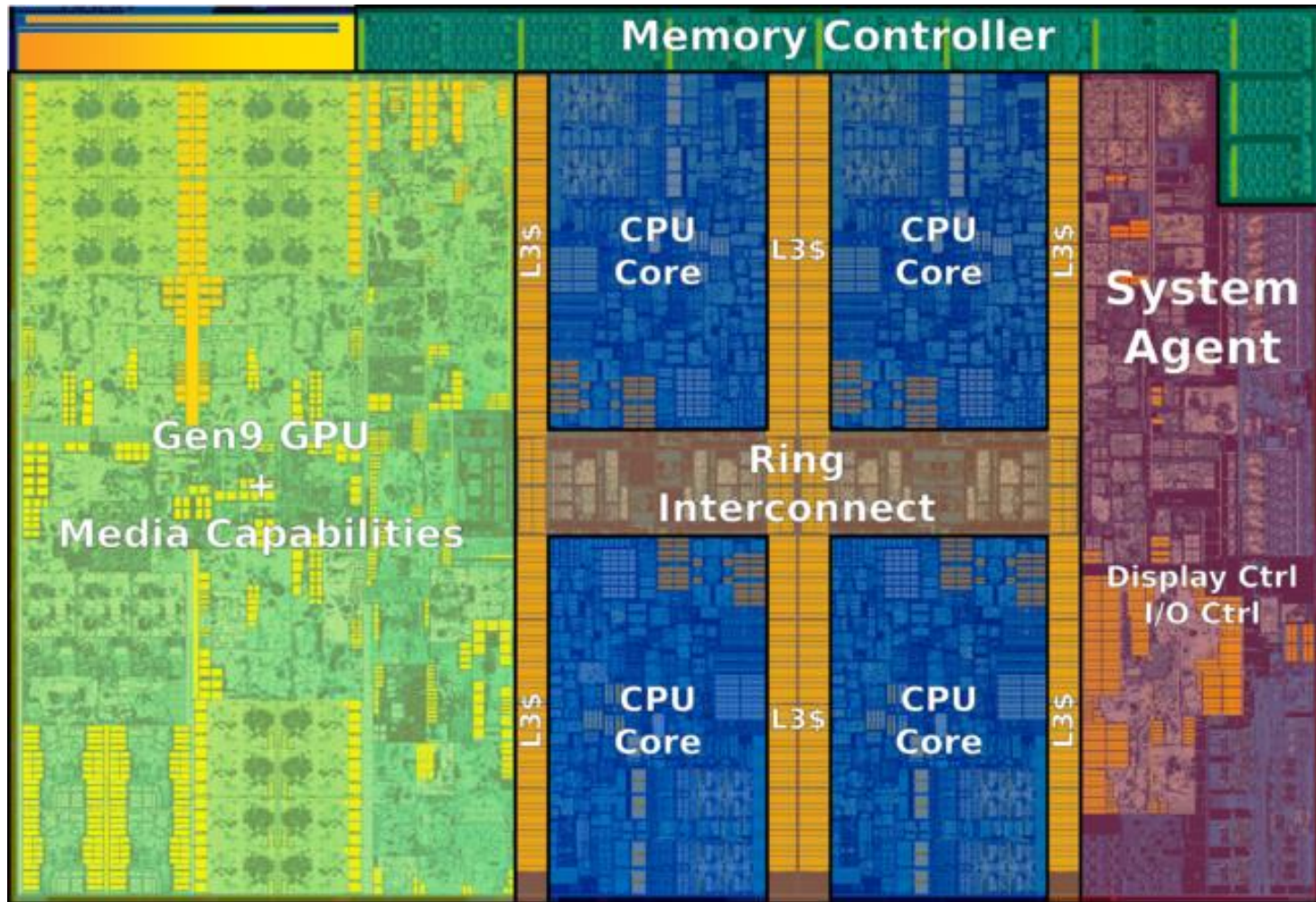
Semiconductor Device Testing

- Functional vs. Structural
- Functional tests ensure logic is correct and the circuit perform all its intended functions within design spec
 - Based on high level model
 - Low fault coverage
- Structural
 - Target is HVM
 - Defect oriented
 - Test vectors are designed based on gate level model
 - To be effective, DFT design technics are used to make the circuit controllable & observable

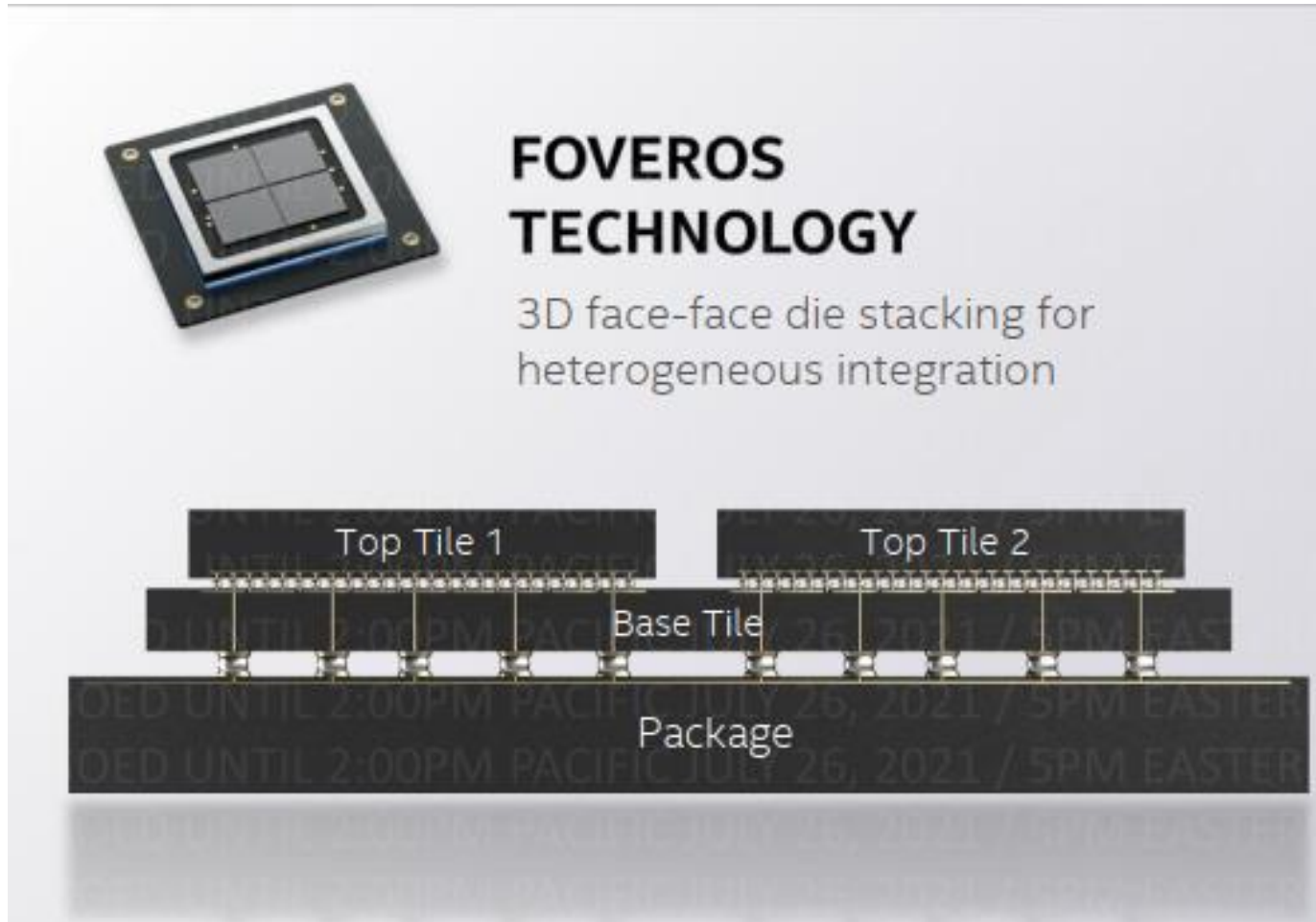
DFX overview

- DFX – DFT, DFM, DFD, DFV
- DFT: Design for Test or testability
 - Primarily for HVM
 - Goal is to enable low cost, high coverage, predictable production test
- DFM: Design for Manufacturing
 - HVM
 - Yield/process improvement
- DFD: Design for Debug
 - Support post-silicon debug activity, observability
 - Support for survivability - defeaturing
- DFV: Design for validation
 - System level validation stress

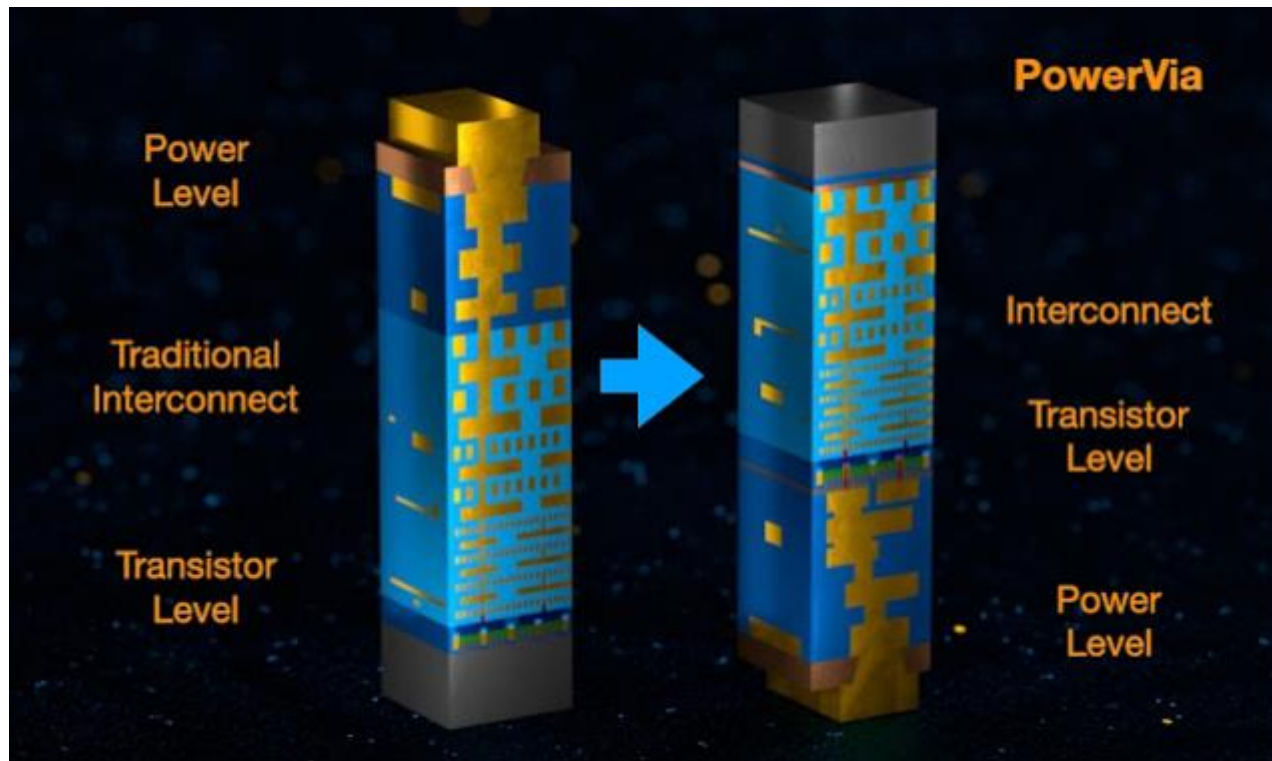
Skylake example



Disaggregation example



PowerVia



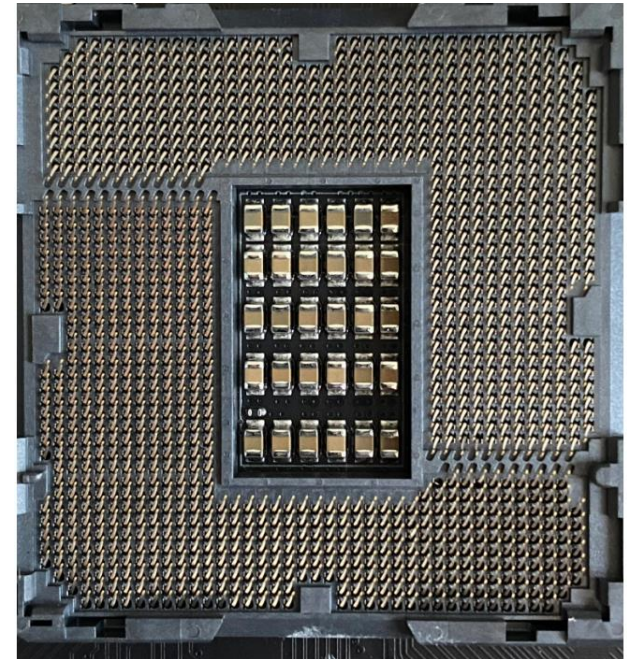
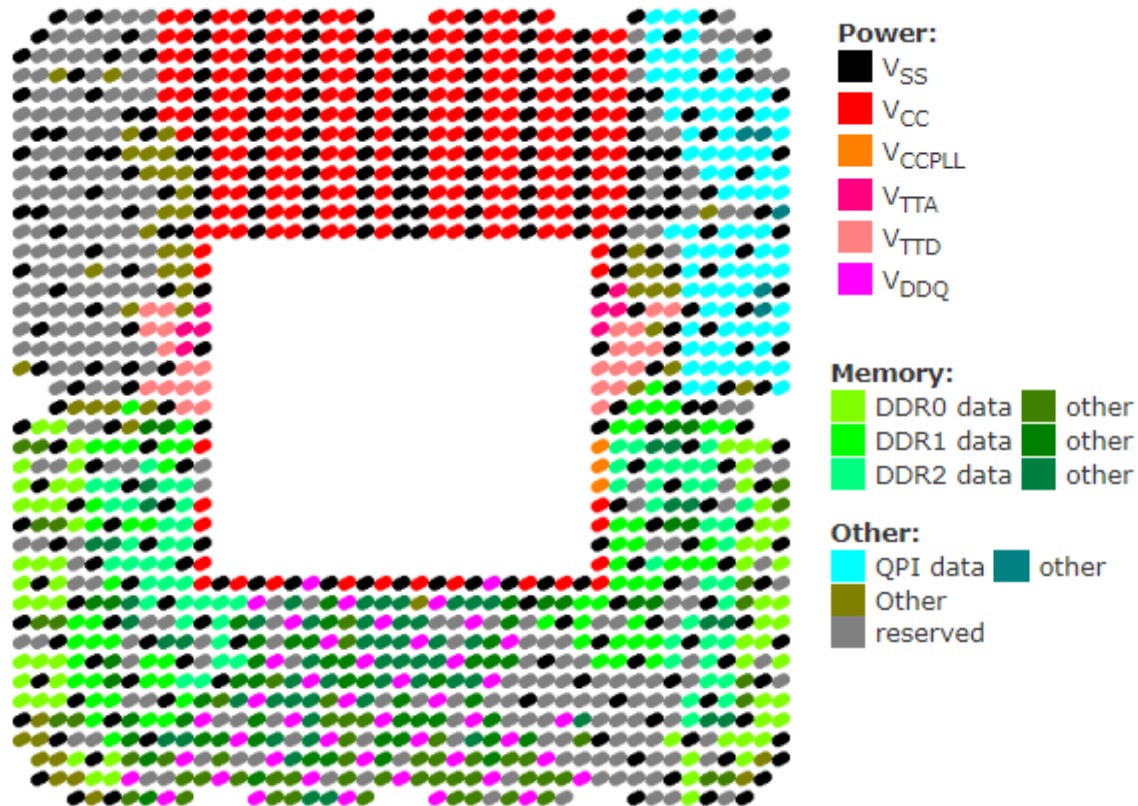
DFX overview

- To verify a complex design
 - Are there tests that can detect all faults
 - Can the test development be kept in check
 - Cost & development time
 - What about test time
- It is a concept where a module other than main portion of a chip is added to test its functionality
- Test coverage
 - DPM: PPV escape, defect per million, indication of outgoing quality
 - High coverage is needed to meet quality while minimizing cost
 - Arrays : BIST
 - Logic: SA vs. at Speed
- Component level DFT
 - Digital, memory & analog/mixed signal circuits
 - Commonly used for digital: SCAN, BIST
- System level DFT (board level or multi-chip package)
 - Boundary scan
 - Analog test bus

DFX overview - why

- Chip internal nodes difficult to control & observe
 - Limited number of external pins
 - Can only be tested with access to internal structures
 - DFT enables better debug control & exposes internal circuitry
- IO are limited in speed, can't exercise the part fully
 - Need to test internal at speed
 - DFT makes at speed test possible
- Traditional functional tests aren't sustainable
 - High speed tester are expensive
 - Test time not practical
 - DFT improves test efficiency & reduce cost

DFX overview - why



Fault Models

- Defect vs. fault vs. error
- Fault models characterize the behavior of a defective circuit
 - Make it possible to analyze defects
 - Test patterns are needed to detect a specific type of defect
 - Fault models enable pattern development and quantify the effectiveness
- Fault modeling levels:
 - Component level: actual transistor faults
 - Functional: design is a black box, IO faults only
 - Gate level: circuit elements & interconnects

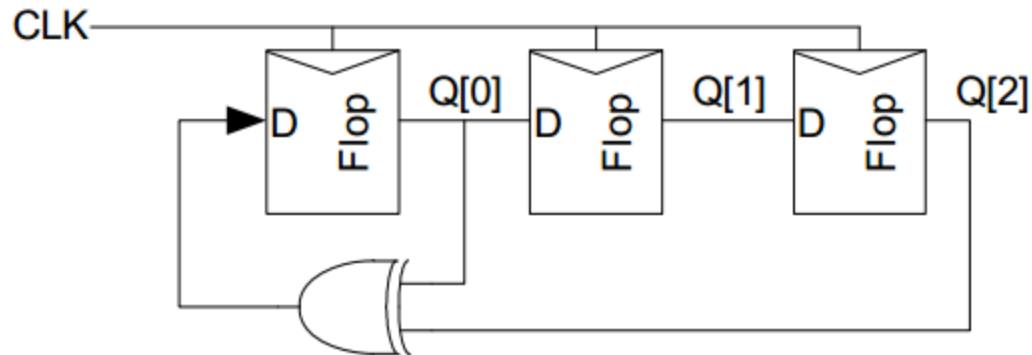
Fault Models

- **Stuck-at fault (SAF)**
 - A node that is permanently low or high regardless of controlling inputs
 - Assumption:
 - only a single line contains a fault at any given time
 - Controllable node from primary inputs
 - Can be propagated to the output
- **Bridging fault**
 - Short between two signals
 - Effect is logic AND or OR of both signals
 - Dominant logic 1 called OR bridge
 - Dominant logic 0 called AND bridge
- **Open fault (SOF)**
 - Output disconnected from input
- **Transition fault**
 - Models slow to rise/fall timing issues
- **Delay fault**
 - Combinational logic delay

BIST

- Built-In Self Test
 - Need at speed test solution
 - Need for reduced test cost
 - On chip pattern generation & response checking
 - Random seed to generate vectors
 - Signature to compare to expected
 - Use SCAN to control, possibly programmable (PBIST)
 - part of power sequence even
- Test vector generation
 - LFSR (Linear feedback shift register)
 - PRBS (pseudorandom binary sequence)

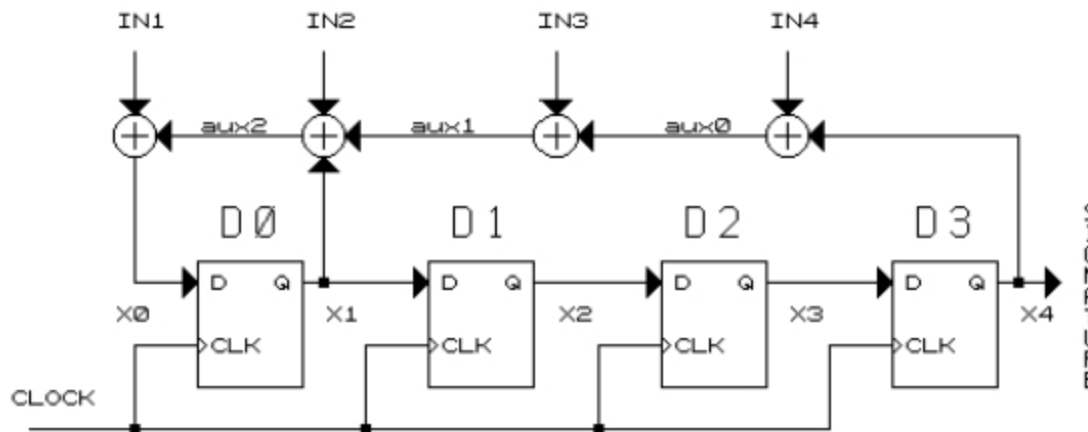
BIST - PRBS



0	111
1	110
2	101
3	010
4	100
5	001
6	011
7	111

BIST - Signature

- SISR (serial input signature register)
 - XOR the LFSR input with output of the circuit under test
 - Basically data compression
 - Possibility of aliasing – not likely if SISR & input sequence is long
- MISR (multiple input signature register)

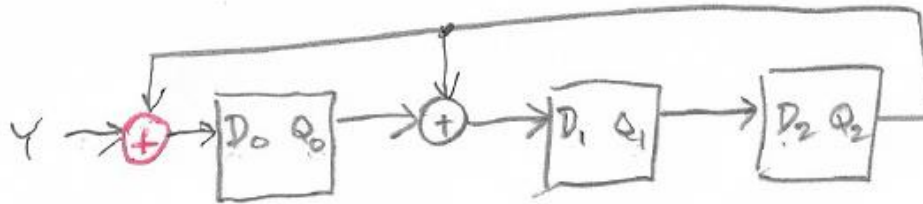


e.g.

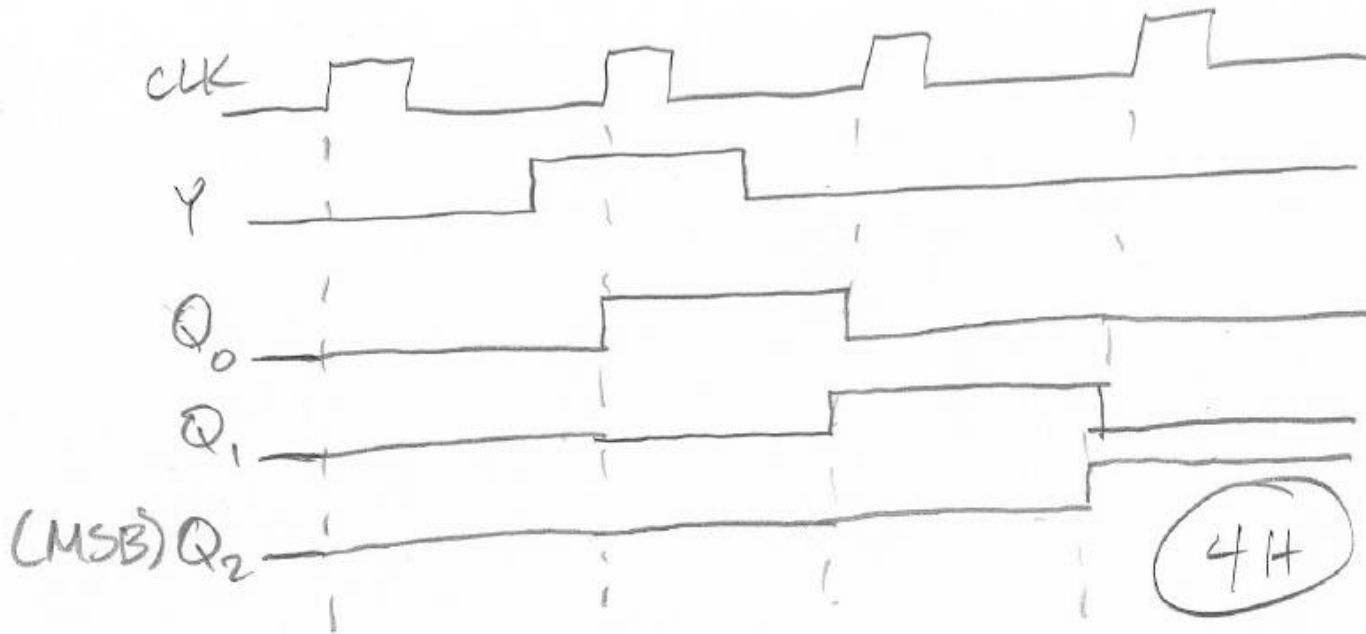


AB	Y
00	0
01	1
10	0
11	0

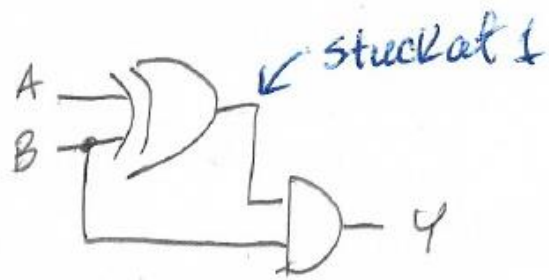
order of input
↓



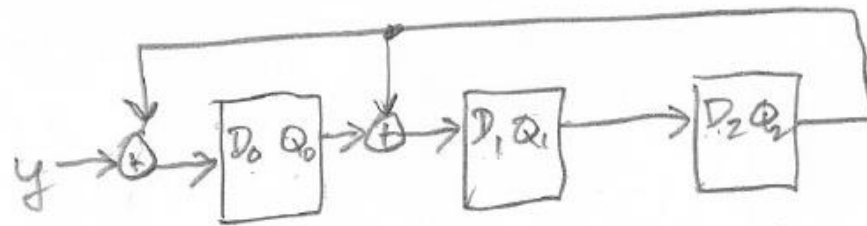
initialize
 $Q_0 = Q_1 = Q_2 = 0$



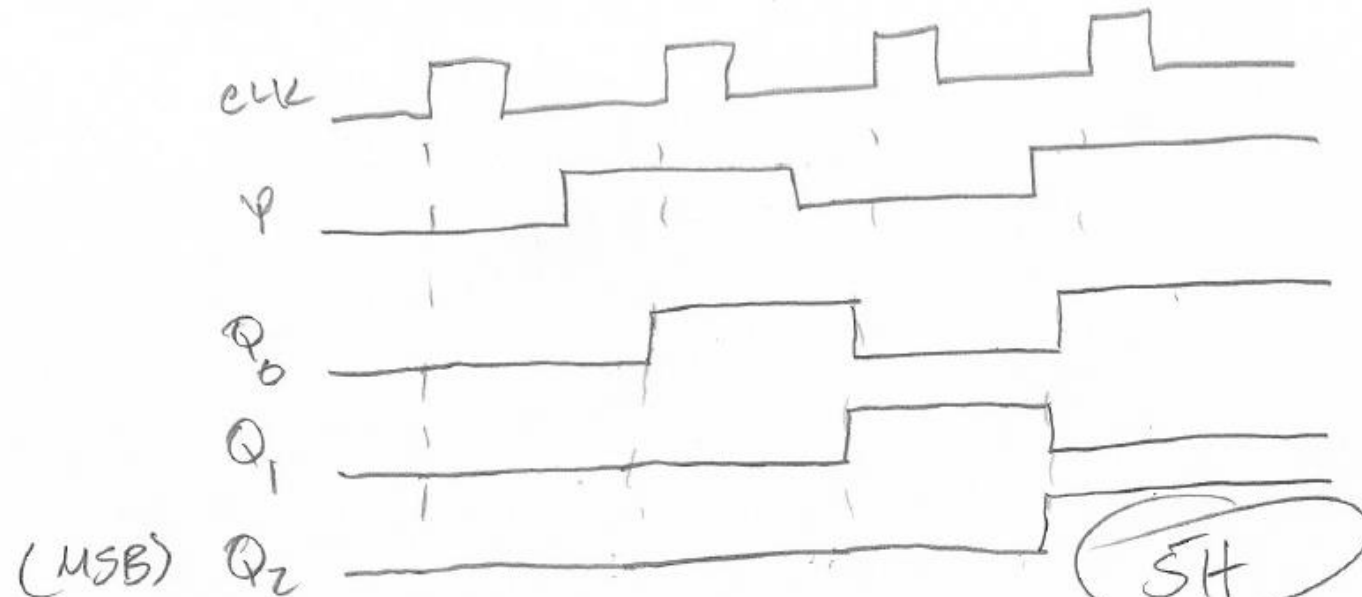
(44) signature



A	B	Y
0	0	0
0	1	1
1	0	0
1	1	1



initialize: $Q_0 = Q_1 = Q_2 = 0$



5H

signature

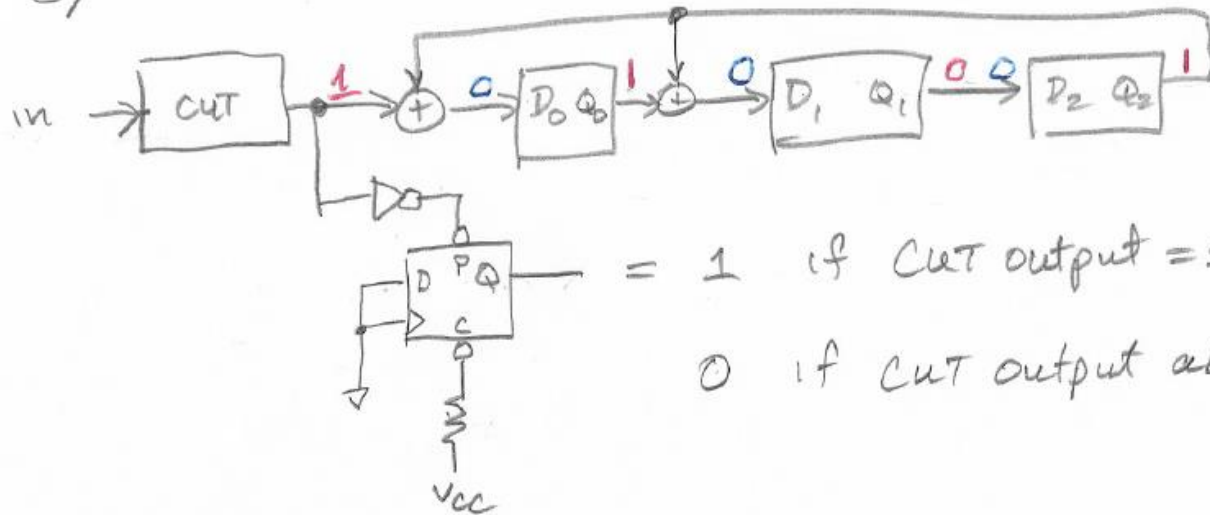
Comments

- 1) signature registers should be large $\Rightarrow$ large n

signature reg	possible signatures
3	$2^3 = 8$
16	$2^{16} = 64K$

- 2) is 00000...0 a valid signature?

YES



Initialize
 $Q_0 = Q_1 = Q_2 = 0$

= 1 if CUT output = 1 at least once
0 if CUT output always 0

Scanout

- Take a snapshot of the state
- Observe only

